

GUILLERMO ESPAÑA JIMÉNEZ

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EXPERIENCE

Siemens Mobility - Rail Automation S.A.U.	Dec 2025 - Present
Software Engineer Internship	Tres Cantos, Spain
- Designed and implemented a Java and Groovy modular diagnostic architecture for early anomaly detection during artifact generation, reducing downstream correction time by 90% and streamlining the SDLC. - Led the refactoring of +80kloc of legacy code following SOLID principles to achieve a Triple A SonarQube quality rating, mitigating security vulnerabilities and enhancing reliability for critical rail systems. - Standardized quality governance by auditing SonarQube rulesets and automating static analysis integration within Gradle/IntelliJ, ensuring architectural consistency and code validation across engineering teams. - Developed automated verification suites achieving 80% code coverage (JUnit, Mockito, JaCoCo), integrating AI-driven coding assistants (GitHub Copilot/Claude Opus) to accelerate the delivery and coverage. - Collaborated in an international Scrum environment, managing agile workflows via Jira/GitLab and authoring technical documentation to facilitate traceability and knowledge transfer within a DevSecOps culture.	

EDUCATION

Technical University of Madrid (UPM)	Expected Aug 2026
Bachelor of Software Engineering	Madrid, Spain
Relevant Coursework: Data Structures and Algorithms, Statistics, Linear Algebra & Discrete Mathematics, Software Architecture, Computer Architecture, Operating Systems, Software Verification & Validation. Distinction for Excellence in English for Professional and Academic Communication (2025).	

ADDITIONAL EDUCATION

Full-Stack Web Engineering (UPM)	Nov 2025
- Architected and deployed full-stack applications using the MERN stack (MongoDB, Express, React, Node.js), developing back-end services using APIs REST focused on scalability, security, and efficient data persistence. - Implemented automated CI/CD pipelines leveraging GitHub Workflows and Jenkins to enforce code quality and streamlined deployment cycles.	
Leadership Development (UPM Leadership and Technology Center)	Feb 2024
Completed training in team leadership, developing core competencies in task coordination, conflict resolution, effective communication and proactivity in engineering environments.	
Global Security Analysis (ONGAWA Engineering for Human Development)	Oct 2023, 2024
Analyzed global technology trends over two years to evaluate socio-economic impacts and the feasibility of engineering solutions for humanitarian and environmental challenges.	

VOLUNTEERING

International Student Mentor (UPM)	2024 - 2025
Mentored students in Computer Networks and OS subjects, utilizing English as the primary language for instruction.	

TECHNICAL SKILLS

Languages: Java (Proficient), Groovy, C/C++, Rust, Python, SQL, C#, TypeScript.
Technologies & Frameworks: React.js, Node.js, Express.js, NumPy, PyTorch, Pandas, Hibernate.
DevOps & Tools: Linux, Bash, Git/GitLab, Jenkins, Azure DevOps, Gradle, SonarQube.
Concepts: Distributed Systems, Microservices, Object Oriented Programming, NoSQL, Database Normalization, Compilers, Computer Networks, Virtualization, Encryption, AI/ML, Neural Networks, APIs, CI/CD, Software Testing.

LANGUAGES

Spanish: Native
English: Advanced (C1)
Japanese, French: Conversational